Complete EDA Checklist

✅ 1. Understand the Dataset

- `df.head()`, `df.tail()` → Preview data
- `df.info()` → Data types, non-null counts
- `df.shape` → Rows and columns
- `df.columns` → Column names
- `df.describe()` → Summary stats for numeric columns

✅ 2. Check for Missing Values

- `df.isnull().sum()` → How many missing in each column?
- `df.isnull().mean()` → Proportion missing (helpful for thresholds)
- Visuals:
  - `sns.heatmap(df.isnull(), cbar=False)`
  - `missingno.matrix(df)` or `missingno.bar(df)`

✅ 3. Check for Duplicates

- `df.duplicated().sum()`
- `df[df.duplicated()]`

✅ 4. Univariate Analysis (Single Variable)

📌 Categorical Features:

- `df['column'].value_counts()`
- Plot: `sns.countplot(x='column', data=df)`

📌 Numerical Features:

- `df['column'].hist()`, `sns.histplot()`, `sns.boxplot()`
- Check:
  - Distribution shape (normal/skewed?)
  - Outliers
  - Mean/Median comparisons

✅ 5. Bivariate Analysis (Between Variables)

📌 Cat vs Cat:

- `pd.crosstab(df['cat1'], df['cat2'])`
- Plot: `sns.countplot(x='cat1', hue='cat2')`, `sns.heatmap()`

📌 Cat vs Num:

- `df.groupby('cat')['num'].describe()`
- Plot: `sns.boxplot(x='cat', y='num')`, `sns.violinplot()`

📌 **Num vs Num:**

- `df[['num1', 'num2']].corr()`
- Plot: `sns.scatterplot(x='num1', y='num2')`, `sns.regplot()`

✅ 6. Outlier Detection

- Use boxplots: `sns.boxplot(x=df['num'])`
- Use z-score or IQR methods for flagging
- Optional: `df.describe(percentiles=[.01, .25, .5, .75, .99])`

✅ 7. Correlation Analysis

- `df.corr(numeric_only=True)`
- Plot: `sns.heatmap(df.corr(), annot=True, cmap='coolwarm')`
- Helps find redundant or highly related features

✅ 8. Feature Relationships & Distribution

- `sns.pairplot(df)` (for small datasets)
- PCA/t-SNE/UMAP if doing dimensionality reduction (advanced)

✅ 9. Target Variable Analysis (if supervised task)

- If classification: Check class balance → `df['target'].value_counts()`
- If regression: Distribution of target + scatterplots with features

✅ 10. Data Cleaning Steps

- Handling missing values (drop, impute)
- Removing or treating outliers
- Encoding categoricals
- Scaling numerical features (if needed later)